

Use Ratio Reasoning

Lesson 3-6

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$2/3 = 8/12$$

Proportion

A math sentence saying two ratios are equal.

$$2/3 = 8/12 \rightarrow 2 \times 12 = 24 \text{ and } 3 \times 8 = 24 \checkmark$$

Cross-multiply

Multiplying across two ratios to check if they are equal.

$$5:8 \times 3 \rightarrow 15:24$$

Scale

To multiply or divide both parts of a ratio by the same number.

$1/2$ and $3/6$ and $5/10$ all equal the same amount —
half

Equivalent

Having the same value.

$$60 \text{ miles in } 3 \text{ hours} \rightarrow 20 \text{ miles per } 1 \text{ hour}$$

Unit rate

A rate for just 1 of something. You find it by dividing.

Key Ideas & Notes

- Chef Academy received a huge catering order — a banquet for 120 guests!
- Chef Reyes's original appetizer recipe serves 8 people and calls for 5 cups of diced tomatoes and 3 cups of mozzarella.
- The students need to use ratio reasoning to scale the recipe so every guest gets the same delicious flavor.
- Chef Reyes wrote several pairs of ratios on the board. Sort them: which pairs are equivalent ratios (proportions) and which are NOT equivalent?

Think About It

- How many times bigger is 120 compared to 8?
- What operation would help you scale both ingredients?
- What stays the same about the recipe even when the amounts change?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Chef Reyes's recipe serves 8 people, but the banquet has 120 guests. How many times bigger is 120 than 8, and how does that scale factor help you adjust the tomatoes and mozzarella?

Level 1 support

Start here: Find the scale factor ($120 \div 8 = 15$), then multiply each ingredient by it.

120 guests is ____ times bigger than 8 because 120 divided by 8 equals ____.

120 invitados es ____ veces más que 8 porque 120 entre 8 es ____.

So I scale by ____, which keeps the recipe in ____.

Entonces escalo por ____, lo que mantiene la receta en ____.

Word bank: **proportion** **scale** **equivalent** **unit rate** **ratio**

Listen for: Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).

Level 2

What stays the same about the recipe even when all the amounts get bigger? Justify why scaling keeps the flavor the same.

- The ____ between tomatoes and mozzarella stays the same because ____.
- Scaling keeps the flavor because ____.

EXPLORE

You sorted ratio pairs into equivalent and not equivalent. How does ratio reasoning, like cross-multiplying, prove that 4:7 and 12:21 are equivalent?

Level 1 support

Start here: Cross-multiplying gives equal products when two ratios form a true proportion.

I reasoned that 4:7 and 12:21 are ____ because ____.

Razoné que 4:7 y 12:21 son ____ porque ____.

Cross-multiplying gives ____ and ____, which are ____.

La multiplicación cruzada da ____ y ____, que son ____.

Word bank: proportion equivalent scale cross-multiply ratio

Listen for: Students show $4 \times 21 = 84$ and $7 \times 12 = 84$ (or that 4:7 scales by 3 to 12:21), proving the ratios are equivalent.

Level 2

Two ratios cross-multiply to UNEqual products. Explain what that tells you and give an example you tested.

- If the cross products are not equal, then ____.
- An example I tested is ____ because ____.

CONNECT

Where might ratio reasoning help you make a smart choice in real life, like shopping or cooking?

Level 1 support

Start here: Scaling recipes, comparing prices, and mixing ingredients all use ratio reasoning.

Ratio reasoning helps me when ____.

El razonamiento con razones me ayuda cuando ____.

I used the ratio to decide ____.

Usé la razón para decidir ____.

Word bank: **proportion** **scale** **equivalent** **unit rate** **ratio**

Listen for: Students name a real use (scaling a recipe, comparing deals, mixing paint or fuel) and explain how keeping the ratio equivalent guides the decision.

Level 2

Describe a situation where ignoring the ratio would cause a real problem, and explain how proportional reasoning fixes it.

- Ignoring the ratio would cause a problem when ____.
- Proportional reasoning fixes it by ____.

Guided Examples

Example 1

A recipe uses 4 cups of broth for every 10 servings. How many cups of broth are needed for 30 servings?

Solution: Scale factor: $30 \div 10 = 3$. Multiply broth by 3: $4 \times 3 = 12$ cups.

Answer: A. 12

Example 2

Are the ratios 6:9 and 2:3 equivalent?

Solution: Divide 6:9 by 3 to get 2:3. Since both ratios simplify to 2:3, they are equivalent.

Answer: A. Yes, both simplify to 2:3

Example 3

If 3 pounds of apples cost \$6, how much do 7 pounds cost?

Solution: Unit rate: $\$6 \div 3 = \2 per pound. For 7 pounds: $\$2 \times 7 = \14 .

Answer: A. \$14

Write About the Math

The Writing Revolution

I can explain my reasoning using the words proportion, scale, equivalent, and unit rate.

1. Kernel Sentence subject + verb

Model: Scale is to multiply or divide both parts of a ratio by the same number.

Escalar es multiplicar o dividir ambas partes de una razón por el mismo número.

Write a kernel sentence about scale. Use a subject and a verb.

Escribe una oración base sobre escalar. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Scale matters in math

Escalar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Scale matters in math because ____.

Escalar importa en matemáticas porque ____.

but
pero

Scale matters in math, but ____.

Escalar importa en matemáticas, pero ____.

so
entonces

Scale matters in math, so ____.

Escalar importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about scale.
Di un hecho verdadero sobre scale.

Scale ____.

Question *Pregunta*

Ask a question about scale.
Haz una pregunta sobre scale.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about scale.
Muestra entusiasmo sobre scale.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with scale.
Dile a un compañero qué hacer con scale.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

120 guests is ____ **times bigger than 8 because 120 divided by 8 equals** ____.

120 invitados es ____ *veces más que 8 porque 120 entre 8 es* ____.

So I scale by ____ , **which keeps the recipe in** ____.

Entonces escalo por ____ , *lo que mantiene la receta en* ____.

I reasoned that 4:7 and 12:21 are ____ **because** ____.

Razoné que 4:7 y 12:21 son ____ *porque* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A chef uses the proportion $\frac{3}{7} = \frac{x}{28}$ to scale a recipe. What is x ?

- A. 12
- B. 14
- C. 9
- D. 21

Show your work:

2. A recipe serves 6 people and uses 4 cups of rice. You need to serve 15 people. Show two different methods to find how much rice you need: (1) using a scale factor and (2) using a unit rate.

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Find Priya's Mistake — find the error, then write the correct reasoning.

Show your work:

Reflect — Exit Ticket

A recipe calls for 6 cups of flour for every 15 cookies. How many cups of flour are needed to make 45 cookies?

- A. 18
- B. 12
- C. 24
- D. 21

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 12 — *Scale factor: $28 \div 7 = 4$. Multiply: $3 \times 4 = 12$. So $x = 12$.*
2. **Try It 2:** Method 1 (scale factor): $15 \div 6 = 2.5$. Multiply rice by 2.5: $4 \times 2.5 = 10$ cups. Method 2 (unit rate): $4 \div 6 = 2/3$ cup per person. For 15 people: $2/3 \times 15 = 10$ cups. Both methods give 10 cups of rice.
3. **Exit Ticket:** A. 18 — *Scale factor: $45 \div 15 = 3$. Multiply flour by 3: $6 \times 3 = 18$ cups of flour.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Scale is to multiply or divide both parts of a ratio by the same number.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).